AxleMetrics

REPORT DATE
2026-01-11

HIGHWAY PAVEMENT DESIGN REPORT

20-Year Design ESAL Analysis

⚠ **Warning: AADT percentages do not sum to 100%**
Current total: 81.0%. Please verify your traffic distribution data.

PROJECT INFORMATION

LOCATION	LENGTH	ANALYST
Kumasi-Accra	300 km	Micah Richards

COMPANY INFORMATION

COMPANY
Kwame Nkrumah University of Science and Technology

PHONE
0201603517

PROJECT SUMMARY

PROJECT NAME
Kumasi-Accra

PREPARED BY
Micah Richards

DESIGN PARAMETERS

BASE YEAR AADT	DESIGN PERIOD	BASE YEAR	DESIGN YEAR
10,000	20 years	2026	2046

DIRECTION DISTRIBUTION	LANE DISTRIBUTION
60%	90%

CALCULATED RESULTS

Vehicle Class	2026 AADT	Dir. AADT	Lane AADT	Growth Rate	Growth Factor	Truck Factor	Design ESALs
7-axle trailers	6,700	4,020	3,618	4.0%	29.78	68.1496	2,679,918,587
6-axle trailers	700	420	378	4.0%	29.78	3889.2761	15,979,018,789
3-axle trailers	500	300	270	4.0%	29.78	77.8329	228,410,598
4-axle trailers	200	120	108	4.0%	29.78	797.4604	936,100,393
						Total 20-Year Design ESALs:	**19,823,000,000**

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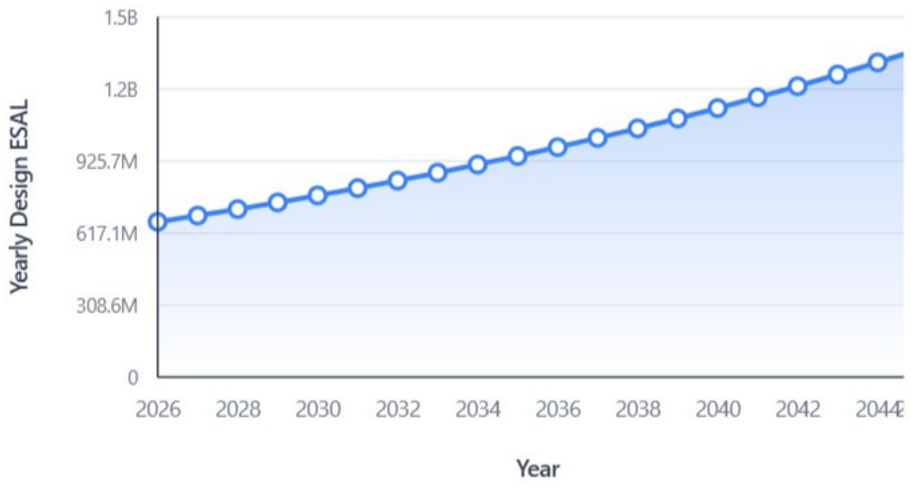
YEARLY DESIGN ESAL VS YEAR (2026 - 2046)



YEARLY ESAL BREAKDOWN

Year	Yearly ESAL	Cumulative ESAL
2026	665,706,094	665,706,094
2027	692,334,337	1,358,040,431
2028	720,027,711	2,078,068,142
2029	748,828,819	2,826,896,962
2030	778,781,972	3,605,678,934
2031	809,933,251	4,415,612,185
2032	842,330,581	5,257,942,766
2033	876,023,804	6,133,966,570
2034	911,064,757	7,045,031,327
2035	947,507,347	7,992,538,674
2036	985,407,641	8,977,946,314
2037	1,024,823,946	10,002,770,261
2038	1,065,816,904	11,068,587,165
2039	1,108,449,580	12,177,036,745
2040	1,152,787,564	13,329,824,309

2041	1,198,899,066	14,528,723,375
2042	1,246,855,029	15,775,578,403
2043	1,296,729,230	17,072,307,633
2044	1,348,598,399	18,420,906,032
2045	1,402,542,335	19,823,448,367

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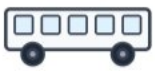
VEHICLE CLASSIFICATION REFERENCE

Passenger Car
PC • 2 axles
2 axles, 4 tires

Small Buses
SB • 2 axles
2 axles

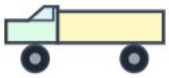


Medium Buses
MB • 2 axles
2 axles

Large Buses (2-axle)
LB2 • 2 axles
2 axles

Large Buses (3-axle)
LB3 • 3 axles
3 axles

Light Trucks
LT • 2 axles
2 axles, pickup/van



Medium Trucks
MT1 • 2 axles
2 axles

Heavy Trucks
HT5 • 3 axles
3+ axles

3-axle trailers
3T1 • 3 axles
Tractor + trailer

4-axle trailers
4T1 • 4 axles
Tractor + trailer

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5-axle trailers
5T1 • 5 axles
Tractor + trailer

6-axle trailers
6T1 • 6 axles
Tractor + trailer

7-axle trailers
7T1 • 7 axles
Tractor + trailer

8-axle trailers
8T1 • 8 axles
Tractor + trailer

9-axle trailers
9T1 • 9 axles
Tractor + trailer

10-axle trailers
10T • 10 axles
Tractor + trailer

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